Final Steps

Before the HMA overlay is placed, the entire surface should be swept thoroughly, then tack-coated immediately before placing the initial lift of HMA. It is recommended to limit traffic on the surface of intermediate HMA lifts until at least 5 inches (125 millimeters) of HMA has been placed over the fractured concrete.

14.4 Rubblization

Rubblization is the process of fracturing the existing PCC pavement in-place into small, interconnected pieces that serve as a base course for a new hot mix asphalt (HMA) overlay. The rubblization process destroys slab action, eliminating the possibility of reflection cracking.

14.4.1 Project Selection and Overview

Whether a pavement is a good candidate for rubblization is a judgment to be made by a qualified engineer. Generally, PCC pavements that require reconstruction or unbonded PCC overlays may also be candidates for rubblization. The Wisconsin Department of Transportation considers the use of rubblization when one or more of the following conditions are met:

- More than 20 percent of the concrete pavement joints needs repair.
- More than 20 percent of the concrete pavement surface has been patched.
- More than 20 percent of the concrete slabs exhibits slab breakup distress.
- More than 20 percent of the project exhibits longitudinal joint distress greater than 4 inches wide.

Unlike other fractured slab techniques, rubblization is suitable for any type of PCC pavement. It is the only method that can be used on continuously reinforced concrete (CRCP). CRCP exhibiting more than five punch-outs per mile should be considered as candidates for rubblization and overlay.

The equipment and processes that have evolved for rubblization were developed for use on relatively thick PCCP slabs constructed over an improved subbase. Thin (less than 6 inches, 150 milimeters) slabs placed directly over poorly drained fine-grained subgrade soils are not good candidates for rubblization.

14.4.2 Design Considerations

The thickness of the HMA overlay should be determined according to the principles outlined in Chapter 6. Falling-weight deflectometer (FWD) testing on rubblized pavements and subsequent back-calculation analysis has consistently shown that the layer coefficient of a rubblized concrete pavement layer is greater than that of a crushed-aggregate base course. Most airfield agencies have structurally characterized a rubblized layer conservatively by treating it as equivalent to a crushed-aggregate base having a CBR of 100. Highway agencies using the 1993 AASHTO design guide have typically characterized

rubblized layers using AASHTO layer coefficients from 0.14 (unstabilized stone base) to 0.30 (stabilized base). The Asphalt Institute recommends using modulus values from the higher end of this range, such as 0.22 to 0.30. The new Mechanistic Empirical Pavement Design Guide (MEPDG) for highways proposes using a modulus value of 150 Kips per square inch (689 Newtons per square meter) for characterizing rubblized concrete pavement. The Asphalt Institute agrees with the value of 150 Kips per square for typical highway pavements. An airfield research project conducted by the Asphalt Institute suggested that the typical back-calculated modulus may be influenced by concrete slab thickness. The following ranges were found:

- Rubblized slabs 6–8 inches (152–203 millimeters) thick: moduli from 100–135 Kips per square inch (689–930 Newtons per square meter)
- Rubblized slabs 8–14 inches (203–356 millimeters) thick: moduli 135–235 Kips per square inch (930–1620 Newtons per square meter)
- Rubblized slabs thicker than 14 inches (356 millimeters): moduli from 235 Kips per square inch–400 Kips per square inch (1620–2758 Newtons per square meter)

The minimum overlay thickness is 5 inches (125 millimeters); the overlay thickness range for interstate highways has been 8–14 inches (203–356 millimeters).

14.4.3 Construction

The procedural steps in the rubblization technique are essentially the same for airfields and highways. These steps are the following:

- Prepare the surface and remove any existing overlays.
- Install a subdrainage system (if necessary).
- Rubblize the concrete pavement.
- Cut off and remove any exposed steel reinforcement.
- Remove exposed joint-sealing material.
- Roll the rubblized concrete pavement.
- Remove and patch any unstable areas.
- Place asphalt HMA leveling course/HMA overlays.

Prepare the Surface

For the rubblization equipment to work properly, any material from a previous overlay must be removed before rubblizing the underlying concrete.

Adjacent pavements that are not to be rubblized must be isolated from the rubblized pavements. This is best accomplished by a full-depth sawcut or by cutting a relief trench with a wheel saw. Either method allows the rubblizing equipment to work up to the project limits without damaging the structural integrity of the pavement that is to remain in place.

Underground structures, utilities, and in-pavement fixtures must be identified as needing protection during the rubblization process. Isolation or removal and replacement may be required during the rubblization process. Provisions must also be made to raise all in-pavement features to match the grade of the new HMA pavement.

Install a Subdrainage System

A key element to a successful rubblization project is to assure that no free water exists immediately under the PCC pavement. This often requires the installation of a properly designed underdrain system. Underdrain systems serve two purposes: before rubblization, a subsurface drainage system dries and stabilizes the subbase or subgrade, and during the service life of the new pavement it prevents water from becoming trapped within the pavement structure.

Any free water trapped in the subbase or subgrade will weaken the subgrade and cushion the impacts from the pavement breakers. This loss of subgrade support reduces the ability of the equipment to properly fracture the concrete. To ensure proper subgrade drainage, many rubblization specifications require drainage to be installed a set period of time (often two weeks) before commencing rubblization.

During the service life of the pavement, the underdrain system must have sufficient capacity and depth to collect and carry the water being discharged from the subgrade, the old permeable base, and the top or "permeable zone" of the rubblized layer (see Figure 14.10). The upper 2 inches of the rubblized concrete is porous, but the remainder of the layer does not appear to be highly permeable. The edge drainage system should be designed to collect and dispose of moisture directly below the interface between the rubblized PCCP and the new HMA layer. The trench depth must also be deep enough to accomplish the intended drainage function. It is recommended that the trench be deep enough to allow the top of the pipe to be located a minimum of 2 inches (5 centimeters) below the bottom of the permeable base material. The filter fabric and trench backfill material must also be designed with sufficient capacity to handle the outflow from the new pavement structure. Erosion of fines into the side-drain system should not pose problems because the base, or rubblized concrete, should not contain erodable fines that might clog the system.

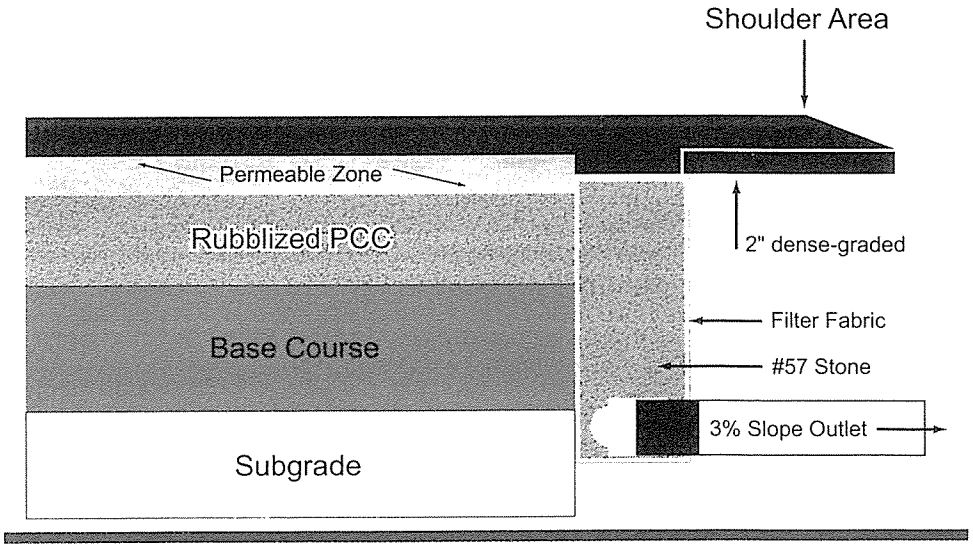


Figure 14.10 Typical Drainage System

Polyethylene tubing used in agriculture and thin "fin-style" drains like those used along highways do not have sufficient strength or capacity to handle the outflow and should not be used for airfield underdrains. Pipe underdrains, such as rigid corrugated PVC with smooth interior walls, are recommended because of their strength, high flow capacity, and configuration for easy maintenance.

Installation of the outlet pipe is critical to the edge-drainage system. It is recommended that a metal or rigid (PVC) nonperforated pipe be used for the outlet pipe because of its strength and capacity. A grade of at least 3 percent is recommended so that the pipe will continue to drain if there is a slight variance of the pipe grade. Because the purpose of subsurface drainage is to remove water as quickly as possible, it is recommended that the outlet spacing be a maximum of 250 feet (75 meters). The edge-drainage design should be coordinated with surface drainage to handle removal of all sources of water.

Rubblize the Concrete Pavement

There are two basic types of rubblization equipment: the Resonant Pavement Breaker (RPB) and the Multihead Breaker (MHB). These two machine types are described below.

Resonant Pavement Breaker

Various models of the resonant pavement breaker exist. All are self-contained, self-propelled units that vary in size and weight. The largest model is the RB-500 (see Figure 14.11), which is a sixth-generation machine developed to rubblize heavy-load PCCP—thicker than 13 inches

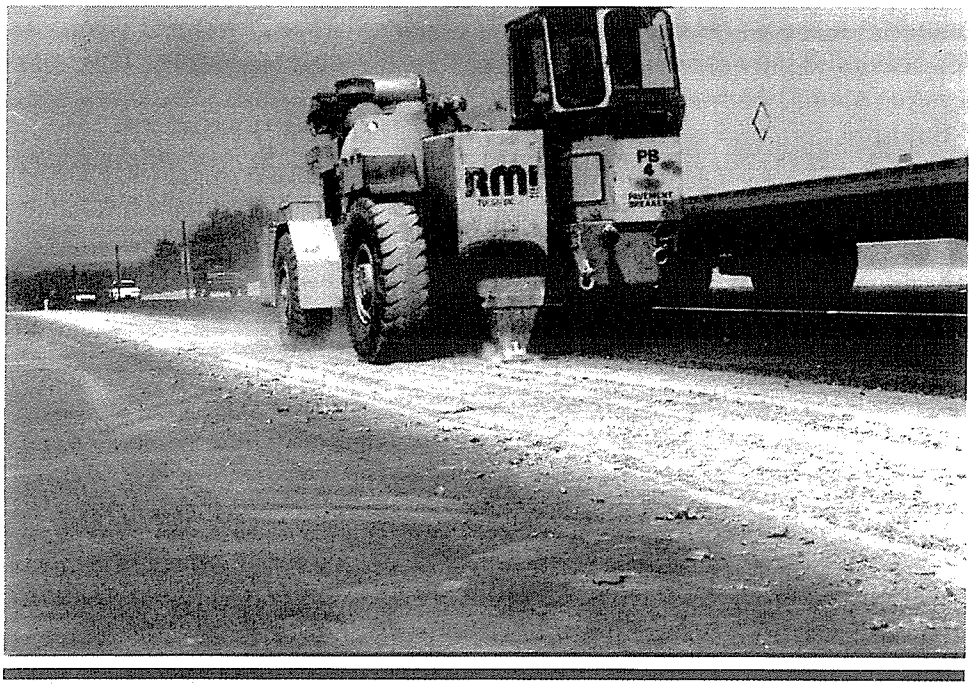


Figure 14.11 Resonant Concrete Pavement Breaker

(330 millimeters). The RB-500 is powered by a 600 horsepower (447 kilowatt) diesel engine.

The concept for resonant rubblization was developed by Ray Gurries as a means to facilitate the removal of reinforced PCC slabs. High-frequency (42–46 Hz), low-amplitude (less than 1 inch; 25 millimeters) blows are generated by an oscillating beam and applied through a steel-breaking shoe. Diagonal cracks are induced through the entire thickness of the slab, while the PCC near the surface is crushed, resulting in smaller particles near the surface. The broken concrete is debonded from any reinforcing steel and remains tightly interlocked.

The RPB rubblizes the PCC pavement in narrow strips as the machine moves forward adjacent to the edge of a previously fractured pavement. It continues with successive passes until it has moved transversely across the width of the pavement. The forward speed of the machine depends on the size of the machine, slab thickness, stiffness of the PCC, and the support provided by the subgrade. The machine is operated at a forward speed of less than 5 miles per hour (8 kilometers per hour). One machine can be expected to rubblize 6,000–8,000 square yards (5000–7000 square meters) per workday shift, with thick airfield pavement work being on the low end of this range. It is not unusual for multiple machines to be employed to expedite the process on major interstate projects.

As an RPB fractures an existing PCC pavement, a small amount of dilation takes place. For extremely wide pavements, such as airport runways and taxiways, relief trenches sometimes have to be cut full depth through the pavement at fixed intervals to allow for horizontal expansion and to reduce the confining pressure at interior locations on these pavements. This is usually not necessary for highway pavements. To remain intact, any existing PCC pavement must be isolated by using a full-depth sawcut to avoid damage. No damage has been observed from operating a resonant breaker adjacent to new HMA placed during shoulder or edge-drain installation.

Multiple-Head Pavement Breaker

The multiple-head pavement breaker is a rubber-tire, self-contained, self-propelled unit (see Figure 14.12) that breaks the PCC using a series of drop hammers. It consists of six pairs of hammers—each weighing 1,200 pounds (544 kilograms) measuring 8 inches (20 centimeters) in width—mounted laterally in pairs, with half the hammers in a forward row and the remainder diagonally offset in a rear row so the breakage is continuous from side to side. Wing units, carrying two 1,500-pound (680 kilograms) hammers, can be added to each side for a total breaking width of up to 13 feet (4 meters). In the 16-hammer configuration, the MHB weighs 57,000 pounds (25,855 kilograms). Each pair of hammers is attached to a separate hydraulic lift cylinder with an adjustable drop height from 60 inches (0–1.5 meters). Each pair operates as an independent unit, and develops 1,000–8,000 foot-pounds of energy, depending on the drop height selected, and cycles at a rate of 30–35 impacts per minute.

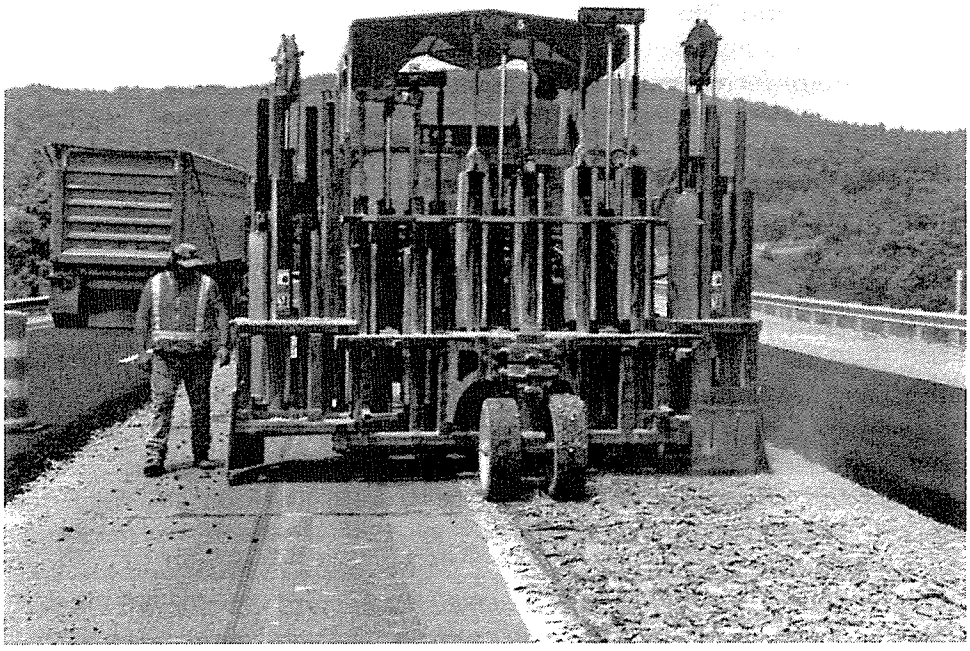


Figure 14.12 Multiple-Head Pavement Breaker

By controlling the individual lift cylinder for each pair of hammers, the MHB can rubblize in widths of 2–13 feet (0.6–4 meters) at a production rate of 500–1500 square yards (418–1254 square meters) per hour. This allows the multiple-head breaker to rubblize lanes that are 12.5 feet (3.8 meters) wide, while traveling at speeds of 0.07–0.20 miles per hour (0.11–0.32 kilometers per hour). Production rates are comparable to the RPB and depend on the strength and thickness of the slab and the underlying subgrade.

Rubblization using an MHB requires a single pass over an area to be rubblized. The tractor or power unit travels on the unbroken slab as it moves forward and tows the breaking unit behind. Because it is a single-pass technique, the MHB does not have to travel on the rubblized surface. Fracturing of the slab is controlled by a set combination of drop height, frequency of the impacts, and the forward speed of the machine, as determined by a test strip/pit.

Part of the process of using an MHB is to roll the broken pavement using a smooth steel-wheel roller onto which a Z-grid pattern has been attached. This breaks any elongated surface particles into a more desirable, cubical shape. Its operation is described later in this section.

Cut Off and Remove Any Exposed Steel Reinforcement

Reinforcing steel or wire mesh that emerges above the surface during rubblization must be removed by cutting with a torch that is flush with the surrounding surface. Otherwise, reinforcing steel, wire mesh, and load-transfer devices should be left in place.

Remove Exposed Joint-Sealing Material

Joint sealant materials such as rubberized asphalt and low modulus silicone are usually loosened during rubblization but often left in place. Any wood joint fillers should be removed before rubblizing the PCC pavement.

Roll the Rubblized Concrete Pavement

The PCC pavement must be rolled after rubblizing. Rolling tightens the surface by seating loose particles as well as by smoothing the surface in preparation for paving. This rolling is not intended to achieve additional densification of the rubblized material or underlying layers. Excessive rolling could reduce particle interlock and cause a weakening of the rubblized layer.

When the rubblization is done with an RPB, the rubblized concrete should be rolled with two to four passes of a smooth tandem vibrating roller (minimum of 10 tons; 9,000 kilograms) for surface smoothing and further alignment of the fractured concrete pieces. The roller should be operated at the highest-frequency setting available at a speed that will produce at least 10 impacts per foot (33 impacts per meter).

When the rubblization is done with a multiple-head breaker, the rubblized concrete should be rolled with two passes of a vibratory steel-drum roller fitted with a special "Z" pattern grid on the drum face (see Figure 14.13). The grid roller should weigh a minimum of 14 tons (12,700 kilograms). This provides a uniform surface and reduces the size of the surface pieces. The grid roller is followed by a 25-ton (22,680 kilograms) pneumatic-tire roller and often a tandem vibratory roller for final seating.

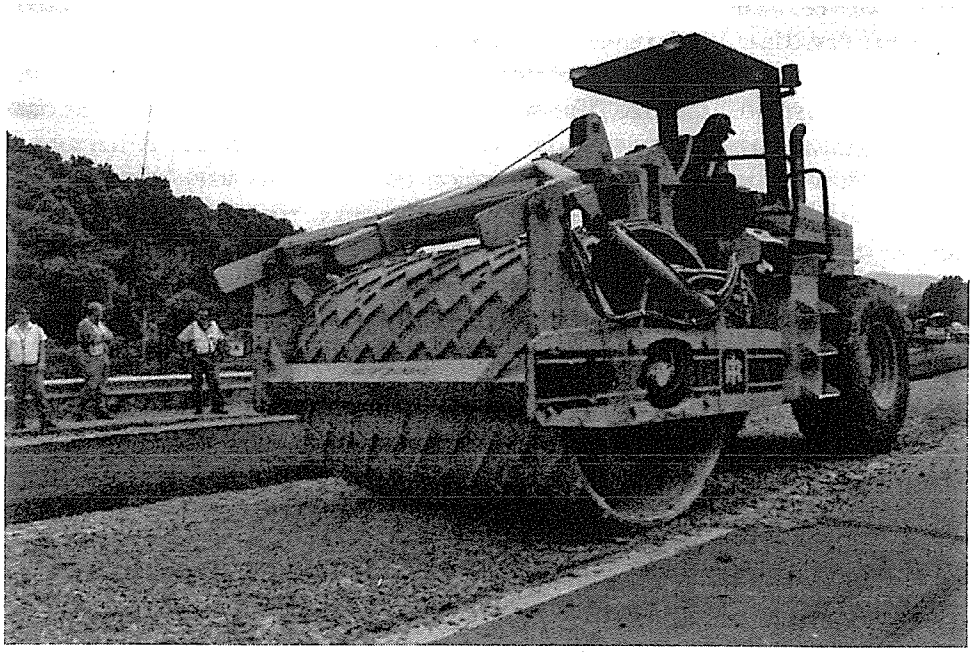


Figure 14.13 Grid Roller Used for Compaction of Rubblized PCCP

Remove and Patch Any Unstable Areas

During rubblization and rolling, any area found to be unstable must be removed and replaced. It is recommended that the section be replaced with a high-quality aggregate base or with HMA.

Place Asphalt HMA Leveling Course/HMA Overlays

Rubblized surfaces cannot be trimmed or fine graded like conventional granular surfaces. Larger pieces will be disturbed by fine-grading equipment, resulting in the loss of particle interlock and an overall reduction in the strength of the rubblized layer. Leveling courses may be necessary to accommodate grade and profile corrections.

Placement of the HMA overlay on a rubblized surface is accomplished according to the procedures discussed in Chapter 8, "Hot Mix Asphalt Paving." Tack coats are not used because of the pickup of the rubblized surface by the construction equipment. Belly-dump trailers and windrow elevators for transferring the HMA into the hopper of the paving machine should not be used to place the first lift of HMA over rubblized PCC. It is suggested that the first lift be placed with a tracked paving machine. It is also recommended that traffic be kept off the surface until the entire HMA overlay has been placed.

14.4.4 Specification Criteria and Inspection

The rubblization process will result in smaller particles near the surface of the pavement; these particles will range from sand size to 3 inches (75 millimeters) to 12–15 inches (300–380 millimeters) at the bottom of the rubblized layer on very thick PCC pavements. The actual size will vary depending on the slab thickness, the type of reinforcement, the subgrade strength, and the equipment used to rubblize. The specifications usually require that the contactor dig a test pit on the first day of production to verify the efficiency of the rubblization process. The specification requirements vary. Airfield projects funded by the Federal Aviation Administration require at least 75 percent of the particles smaller than 3 inches (75 millimeters) at the surface and 12 inches (300 millimeters) in the bottom half. Other highway projects require all the particles to be less than 10 inches (250 millimeters) on the surface.

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CHAPTER 15

SPECIAL USES OF ASPHALT AND HMA

This chapter describes the many special uses of HMA and asphalt. The details of preparing and using cold mix asphalt mixes are covered. Using asphalt in hydraulic applications and for railway track beds is described. Information is provided on how asphalt is used in athletic and recreational facilities. Asphalt roofing is summarized and mitigating pavement noise with selected asphalt mixes is discussed. The special requirements for asphalt for race and test tracks are covered. New applications of asphalt in porous pavements and in warm mix technology are covered.

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15.1 Cold Mix Asphalt

15.1.1 Introduction

Cold mix asphalt preparation is not as complicated as that described for hot mixes in Chapter 7. Generally, placing and compacting does not usually call for equipment as sophisticated as that required for hot mix asphalt. However, this type of pavement construction is no less important, because a large percentage of the nation's roadways have cold mix asphalt bases and even surfaces.

A less-controlled, simplified process is an advantage during in situ cold mix preparation with a distributor and a motor-grader operation. But the same quality control as that found in hot mix asphalt preparation is possible when a cold mix is prepared in a plant-mix operation. However, many plants used for producing cold mixes do not use, or have, all the means for maintaining close uniformity of control. Wherever suitable materials are available, such plant-mixing procedures are employed for high production and low cost.

Cold mix recycling or reworking of existing asphalt pavement or aggregate structures is another important pavement construction alternative. Cold mix recycling is the process by which reclaimed asphalt pavement (RAP), reclaimed aggregate materials (RAM), or both are combined with new asphalt in place or at a central mixing facility to produce cold mix base mixtures. For more complete information on cold mix recycling, see Chapter 13, "Rehabilitation Methods for Asphalt Pavements."

15.1.2 Description

Cold-placed asphalt pavement mixtures (cold mixes) are generally made with emulsified or cutback asphalt. Emulsified asphalt may be anionic or cationic MS and SS grades. Cutback asphalt may be MC or SC grades. Special additives to the emulsions add a High Float (HF) designation to the grading system and will slow the breaking of the emulsion when the reclaimed material has a high percentage of fine material or minus No. 200 (75 microns) sieve. The Asphalt Institute's MS-19, *Asphalt Emulsion Manual* should be consulted for more information.

The aggregate material may be anything from a dense-graded crushed aggregate to a granular soil having a relatively high percentage of dust. The aggregate at the time of mixing might be damp, air-dry, or artificially heated and dried.

Mixing occurs either in the roadway, along the roadway, or in a stationary mixing facility. The resulting mixtures are usually spread and compacted at ambient temperature; however, there are exceptions. Some asphalt mixes

made with MC- and SC-3000 are similar to hot asphalt mixes because they are mixed and placed at temperatures exceeding 200°F (93°C). Also, some emulsion mixtures may be produced at elevated temperatures.

15.1.3 Suitability for Pavement Construction

Cold-laid asphalt mixes may be used for surface, base, or subbase courses if the pavement structure is properly designed. Cold-laid surface courses are suitable for light and medium traffic. When used in the base or subbase, they may be suitable for all types of traffic. Also when used on large jobs, cold-laid paving mixtures reduce construction costs, but not necessarily the strength or quality of the pavement structure. High quality cold mixes, if properly designed and constructed, can provide a structural equivalency equal to that of hot mixes.

Asphalt pavements made by cold-laid methods can be designed and built to meet current traffic requirements. A thicker pavement can be achieved by adding additional aggregate material if required, during in-place mixing or by adding additional pavement courses.

15.1.4 Mix Composition

A wide variety of aggregates and soil-aggregate combinations can be used in cold-mixes. In many cases, material from road cuts, pits, or nearby quarry sites may be economically crushed or used. Often the entire output of a rock crusher less than a certain maximum size is used.

Suitable local materials also include granular soils having up to 20 percent passing the No. 200 (75 microns) sieve, as well as clean sands and gravel.

Granular materials that are mostly fine aggregate and have a relatively high percentage of mineral dust may cause mixing, aeration, and compaction problems.

Well-graded aggregates are always desirable for any course of the asphalt pavement structure. However, poorly graded and gap-graded aggregates can be adequate for base courses when they are properly prepared by road mixing methods and procedures.

15.1.5 Asphalt

The type and grade of asphalt selected for use in cold-laid pavement construction depends a great deal on the aggregate, mixing method, and climatic conditions. Recommended asphalts for different types of cold-laid pavement mixtures are given in Table 15.1.

Mixtures prepared at a stationary mixing facility generally permit a more viscous grade of asphalt than that used in the road-mixing method. Cool weather requires a lower-viscosity grade and permits the selection of a

Table 15.1
Guide for Uses for Asphalts in Cold Mix

Type of Construction	Emulsified Asphalts													Cutback Asphalts						
	Anionic									Cationic				Medium Curing (MC)				Slow Curing (SC)		
	MS-1	MS-2	MS-2h	HFMS-1	HFMS-2	HFMS-2h	HFMS-2s	SS-1	SS-1h	CMS-2	CMS-2h	CSS-1	CSS-1h	70	250	800	3000	250	800	3000
Cold-Plant Mix																				
Pavement Base & Surface																				
Open-Graded Aggregate	X	X	X	X	X	X				X	X					X				
Dense-Graded Aggregate							X	X	X						X	X	X	X	X	X
Patching, Immediate Use								X	X			X	X		X	X			X	
Patching, Stockpile												X	X		X	X		X	X	
Mixed-in-Place (Road Mix)																				
Pavement Base & Surfaces																				
Open-Graded Aggregate	X	X	X	X	X	X	X			X	X					X	X		X	X
Dense-Graded Aggregate							X	X	X			X	X		X	X		X	X	
Sand							X	X	X			X	X	X	X	X				
Sandy Soil							X	X	X			X	X		X	X				
Patching, Immediate Use							X	X	X			X	X		X	X			X	
Patching, Stockpile															X	X		X	X	

faster curing or setting asphalt than what might be chosen for warmer or hot weather mixing conditions.

Mixtures prepared with emulsified asphalt do not require aggregates to be as dry as those used in mixtures prepared with cutback asphalt. Such emulsified-asphalt mixtures reduce the amount of manipulation or extent of air drying required before mixing the aggregate with asphalt.

Cold-laid asphalt mixtures prepared for maintenance use and stockpiled over a period of time are usually prepared with a lower-viscosity grade of MC or SC cutback asphalt.

15.1.6 Types of Mixes

Open-Graded Mixes

A great variety of aggregate gradations have been used for open-graded cold mixes. Most of the research and development work on these mixes has centered in the northwestern states, where they have been successfully used as bases and surface courses on many miles of federal, state, and county highways and heavy-duty logging roads. Aggregate gradation and quality requirements vary somewhat, but most of them closely approximate those listed in Table 15.2. Some trial projects used aggregates with up to 20 percent passing the No. 8 (2.36 millimeters) sieve and 5 percent passing the No. 200 (75 microns) sieve and using CMS-2s emulsified asphalt. (Some

Table 15.2
Aggregates for Open-Graded Emulsion Mixes

Sieve Size	Base	Surface	
		Medium	Fine
1 ½ in (38.1 mm)	100		
1 in (25.0 mm)	95–100	100	
¾ in (19.0 mm)		90–100	
½ in (12.5 mm)	25–60		100
⅜ in (9.5mm)		20–55	85–100
No. 4 (4.75mm)	0–10	0–10	
No. 8 (2.36 mm)	0–5	0–5	
No. 16 (1.18 mm)			0–5
No. 200 (75 µm)	0–2	0–2	0–2
Los Angeles Abrasion Loss @500 Rev. (ASTM C 131)	40 max.	40 max.	40 max.
Percent Crushed Faces	65 min.	65 min.	65 min.

Emulsified Asphalt Grades MS-2, MS-2h, HFMS-2, HFMS-2h, HFMS-2s, CMS-2, or CMS-2h

user agencies specify an additional cationic mixing grade designated CMS-2s. The CMS-2s is used for sand and silty sand mixes; it contains more solvent than standard CMS grades). Their performance as high-quality base and surface courses has been variable. Generally, gradations with more than 15 percent passing the No. 8 (2.36 millimeters) sieve do not have sufficient voids within the aggregate structure to provide free drainage.

Dense-Graded Mixes

Dense-graded aggregate mixtures are graded from the maximum size down to and including material passing the No. 200 (75 microns) sieve. They embrace a wide variety of aggregate types and gradations and, similarly, can be used for the full range of base and surface pavement types, depending on aggregate quality and equipment. Substantial savings can be realized when locally available, unprocessed aggregates are used on lightly traveled roads and in bases for heavy-duty pavements.

Samples of all aggregates intended for use in the mixture(s) should be submitted for testing. Recommended aggregate gradation and quality requirements for dense-graded mixes are shown in Table 15.3. For dense-graded mixes, the recommended asphalt grades are shown in Table 15.1.

Table 15.3
Aggregates for Emulsified Dense-Graded Asphalt Mixtures

Sieve Size USA Standard/ Alternative	Semi-Processed Crusher, Pit or Bank Run	Processed, Dense-Graded Asphalt Mixtures				
2 in. (50 mm)		100				
1 ½ in (37.5 mm)	100	90–100	100			
1 in (25.0 mm)	80–100		90–100	100		
¾ in (19.0 mm)		60–80		90–100	100	
½ in (12.5 mm)			60–80		90–100	100
⅜ in (9.5 mm)				60–80		90–100
No. 4 (4.75 mm)	25–85	20–55	25–60	35–65	45–70	60–80
No. 8 (2.36 mm)		10–40	15–45	20–50	25–55	35–65
No. 16 (1.18 mm)						
No. 30 (600 µm)						
No. 50 (300 µm)		6–16	3–18	3–20	5–20	6–25
No. 100 (150 µm)						
No. 200 (75 µm)	3–15	0–5	1–7	2–8	2–9	2–10
Sand Equivalent, Percent	30 min.	35 min.	35 min.	35 min.	35 min.	35 min.
Los Angeles Abrasion @ 500 Rev.	40 max.	40 max.	40 max.	40 max.	40 max.	40 max.
Percent Crushed Faces	65 min.	65 min.	65 min.	65 min.	65 min.	65 min.

Sand Mixes

Except for aggregate gradation, the same basic principles apply for production of sand-emulsion plant mixes as for dense-graded, coarse-aggregate emulsion mixes. Sand mixes may be used for either base or surface courses.

The addition of 1 to 2 percent Portland cement will aid in the development of early initial strength. Thorough mixing is essential to uniform distribution of the cement throughout the mixture.

A wide variety of fine aggregates throughout the country have produced satisfactory results. Blending of two or more aggregates may be necessary to produce the desirable mix characteristics. Typical gradations are listed in Table 15.4.

The emulsified-asphalt content usually varies from 6 to 10 percent. Emulsion grades SS-1, SS-1h, CSS-1, CSS-lh, or HFMS-2s may be used, along with a laboratory-determined amount of mixing water added to the sand.

15.1.7 Proportioning Materials

There is no universally accepted asphalt-emulsion-aggregate mix design method for dense or open-graded cold mixes. Nearly all mix design methods are modifications of the Hveem (ASTM D 1560 and 1561 and AASHTO T 246 and 247) or Marshall (ASTM D 1559 or AASHTO T 245) test methods. When laboratory equipment is not available, formulas may be used for approximating the asphalt content for initial mixing.

For estimating the approximate emulsified-asphalt content in dense-graded mixes, the following empirical formula may be used:

$$P = (0.05A + 0.1B + 0.5C) \times (0.7)$$

Table 15.4
Aggregate for Emulsified Sand Mixes

	Total Percent Passing		
Sieve Size	Poorly-Graded	Well-Graded	Silty Sands
½ in (12.5 mm)	100	100	100
No. 4 (4.75 mm)	75–100	75–100	75–100
No. 50 (300 µm)	-	15–30	-
No. 100 (150 µm)	-	-	15–65
No. 200 (75 µm)	0–12	5–12	12–20
Sand Equivalent, Percent	30 min.	30 min.	30 min.
Plasticity Index	NP	NP	NP

where

P = percent of total emulsified asphalt by weight of dry aggregate
A = percent of aggregate retained on the No. 8 (2.36 millimeters) sieve
B = percent of aggregate passing No. 8 (2.36 millimeters) and retained on No. 200 (75 microns)
C = percent of aggregate passing No. 200 (75 microns)
(All the above percentages should be expressed as whole numbers.)

When MC or SC cutback asphalt is used, the asphalt content may be estimated by using the following formula:

$$P = 0.02a + 0.07b + 0.15c + 0.20d$$

where

P = percent of asphalt by weight of dry aggregate
a = percent of aggregate retained on the No. 50 (300 microns) sieve
b = percent of aggregate passing No. 50 (300 microns) and retained on the No. 100 (150 microns) sieve
c = percent of aggregate passing No. 100 (150 microns) and retained on the No. 200 (75 microns) sieve
d = percent of aggregate passing the No. 200 (75 microns) sieve

Additional asphalt may be required for absorptive aggregates such as slag, limerock, etc.

Example

Given the following aggregate and construction method, determine the required amount of emulsified asphalt.

Method of construction: blade mixing

Aggregate: relatively nonabsorptive gravel, gradation shown below

Asphalt: SS-1

Sieve size	19.0 mm ¾ in	9.5 mm ⅜ in	4.75 mm No. 4	2.36 mm No. 8	600 μm No. 30	300 μm No. 50	150 μm No. 100	75 μm No. 200
Percent Passing	100	75	59	40	20	14	10	5

Solution:

$$P = [0.05(100–40) + 0.1(40–5) + 0.5(5)] \times 0.7 = 6.3$$

Answer: Use for 6 percent emulsified asphalt by weight of dry aggregate.

15.1.8 Construction Methods

Roadbed Preparations

The roadbed on which cold-mixed material is placed must be shaped and compacted. The surface must be swept with a power broom to remove dirt and other foreign matter. Depending upon the condition or type of base or subgrade, a prime coat of MC-30 or MC-70 may be necessary. In some regions, environmental regulations have restricted the use of MC and RC cutback asphalts. After application, the prime coat is allowed to cure, and after 24 hours, any excess asphalt remaining on the surface is blotted with sand.

Windrows

In several types of road-mix construction, it is necessary to place the aggregate into windrows. Windrow sizers are used to ensure that the correct volume for the desired depth of pavement is placed. Usually, loose material on the unpaved road surface is of too small a quantity and not uniform enough in size for use in the road mix. In such cases, the material is bladed onto the shoulders. If the material from the roadway can be practically used, it is bladed into a windrow and measured. Additional aggregate may be added as necessary to improve the gradation or to provide the required volume.

If two or more materials are combined on the area being surfaced, each is placed in its own sized windrow. These windrows are then bladed back and forth to thoroughly mix them and combine them into a single windrow before asphalt is added.

Before the correct amount of asphalt can be determined, it is necessary to determine the quantity of aggregate in the windrow. Measurements to be made on a sized or uniform windrow are shown in Figure 15.1.

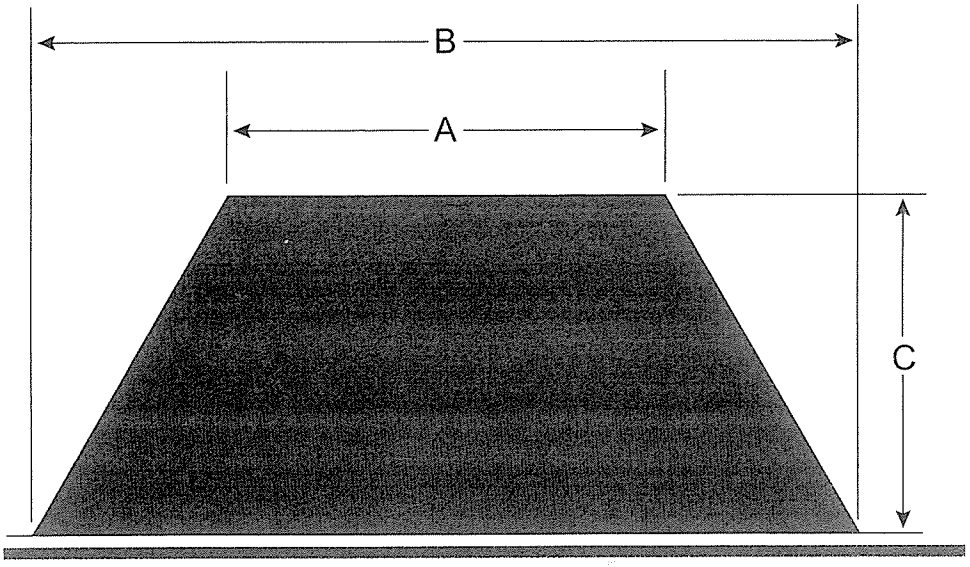


Figure 15.1 Measurements for Determining Windrow Quantities

The quantities are then determined by the following formulas:

$$V = (A+B)\frac{C}{2} \quad \text{(Equation 1)}$$

where

V = volume of the windrow, cubic foot per linear foot (cubic meter per linear meter)

A, B, C = dimensions of the windrow (see Figure 15.1) in feet or meters

$$W_f = W_1 V \quad \text{(Equation 2)}$$

where

W_f = quantity of aggregate (pounds/linear foot; kilograms/linear meter) of windrow

W_1 = loose weight of dry aggregate (pounds/cubic feet (kilogram/cubic meter)

Example

A windrow of dried aggregate 1.64 feet (0.5 meter) high, 1.64 feet (0.5 meter) wide at the top, and 3.28 feet (1.0 meter) wide at the base has a loose dry weight of 93.6 pounds per cubic foot (1500 kilograms per cubic meter). Substituting into Equations 1 and 2:

Customary Units	**Metric Units**
$V = (1.64 + 3.28)\frac{1.64}{2}$	$V = (0.5 + 1.0)\frac{0.5}{2}$
$V = 4.03$ cubic feet per linear foot	$V = 0.375$ cubic meter per linear meter
$W_f = 93.6\ (4.03)$ $W_f = 377$ pounds per linear foot	$W_f = (1500)\ 0.375$ $W_f = 562.5$ kilograms per linear meter

Incorporation of Binder

Asphalt is either applied with a pressure distributor ahead of the mixing process or, in the case of travel mixers, during the mixing process. In any case, close control over the application rate and viscosity is necessary for proper mixing. The rate of application for asphalt along the windrow is determined by the following formula:

$$A = \frac{(W_f \times P)}{(100 \times G)}$$

where

A = application rate of asphalt, gallons per linear foot (liters per linear meter)

W_f = quantity of aggregate, pounds per linear foot (kilograms per linear meter)

P = asphalt content, percent by weight of dry aggregate

G = weight of asphalt, approximately 8 pounds per gallon (1 kilogram per liter)

Example

Aggregate: Weighing 377 pounds per linear foot (562.5 kilograms per linear meter) windrow

Asphalt: MC-250, asphalt content 4 percent by weight of dry aggregate

Customary Units	Metric Units
$A = \frac{377 \times 4}{100 \times 8}$	$A = \frac{(562.5 \times 4)}{(100 \times 1)}$
A = 1.9 gallons per linear foot	A = 22.5 liters per linear meter

Proper viscosity is essential for the asphalt to flow easily through the spray nozzles and adequately coat the aggregate particles. Cutback asphalt, although already fluid, needs some heating to bring it to the proper viscosity for spraying, which is 20–120 centistokes. Proper safety precautions should always be taken when heating cutback asphalt. Mixing should begin immediately, because the viscosity will quickly rise above that recommended for mixing, which is 150–300 centistokes. The volatiles in cutback asphalt keep them fluid long enough for the completion of road mixing. The aggregate temperature should be at least 50°F (10°C) in the shade at mixing time.

Mixing and spraying temperatures for emulsified and cutback asphalt are given in Table 15.5.

Table 15.5
Typical Asphalt Temperatures for Cold Mix Construction

Type and Grade	Asphalt Temperature When the Asphalt Is Metered Through a Mixing Plant	Asphalt Temperature When the Asphalt is Applied to a Windrow Before Mixing
Emulsified Asphalts		
Anionic		
MS-1, MS-2, MS-2h, HFMS-1, HFMS-2, HFMS-2h, HFMS-2s, SS-1, SS-1h	50–160°F (10–70°C)	70–160°F (20–70°C)
Cationic		
CMS-2, CMS-2h, CSS-1, CSS-1h	50–160°F (10–70°C)	70–160°F (20–70°C)
Cutback Asphalts		
MC, SC Asphalt		
250	135–175°F (55–80°C)	105°F+ (40°C +)
800	165–210°F (75–100°C)	135°F+ (55°C +)
3000	180–240°F (80–115°C)	-

1. Temperatures are included in this table as a guide only.
2. The maximum temperature of cutback asphalt shall be below that at which fogging occurs.

15.1.9 Methods of Preparing Mixes

The four techniques for preparing paving mixtures for cold-laid pavements are the following:

- Blade mixing
- Rotary mixing
- Travel-plant mixing
- Stationary mixing facilities

All but stationary plant mixing are considered mixed-in-place techniques. The principal advantage of using mixed-in-place techniques for construction is that they use aggregate already on the site or available from nearby sources. The aggregate and asphalt can be mixed in place with minimal equipment.

The in-place mixing systems provide low-cost techniques for production of cold mixes. It is not possible to achieve 100 percent asphalt coating of the aggregate. Sufficient coating can be achieved to produce a tightly bonded mixture. The mixing procedure selected should be one that minimizes the amount of uncoated aggregate particles.

Stationary mixing facilities provide opportunities for better control of proportioning and mixing. In some cases, such as those mixes made with emulsified asphalt, no heating or drying of the aggregate before mixing is necessary.

Whenever asphalt emulsions are used, the aggregate should be dampened before introducing the emulsion. Dampening slows the breaking of the emulsion. Otherwise, the dry aggregate absorbs the water in the emulsion, which makes coating the aggregate particles very difficult. A good practice is to determine the optimum fluid content (a combination of moisture and emulsion, equal to the optimum moisture of the aggregate) for the mix. That amount of moisture (optimum moisture content minus the emulsion to be added) plus about 2 percent should be applied to the aggregate before introducing the emulsion. This process yields good uniform mixing, with well-coated aggregate particles, and provides the added moisture (which will evaporate during handling and placing of the mix) needed to ensure proper fluid content at the time of compaction.

Blade Mixing

Using motor graders or some type of multiple-blade drag to blend the asphalt and the aggregate usually accomplishes blade mixing (Figure 15.2). The aggregate is first air dried by manipulating it across the roadway with the grader to expose it to the sun and air. After the moisture content has been reduced to 3 percent or less, the windrow is spread uniformly over about half the roadway.

Asphalt is then applied from an asphalt distributor in two or three spraying passes. Immediately following each application, the asphalt and aggregate are partially mixed to leave as little free asphalt as possible on the surface and to prevent pools of asphalt from forming.

After receiving the full predetermined amount of asphalt, the partially mixed material is formed back into a windrow. It is then worked back and

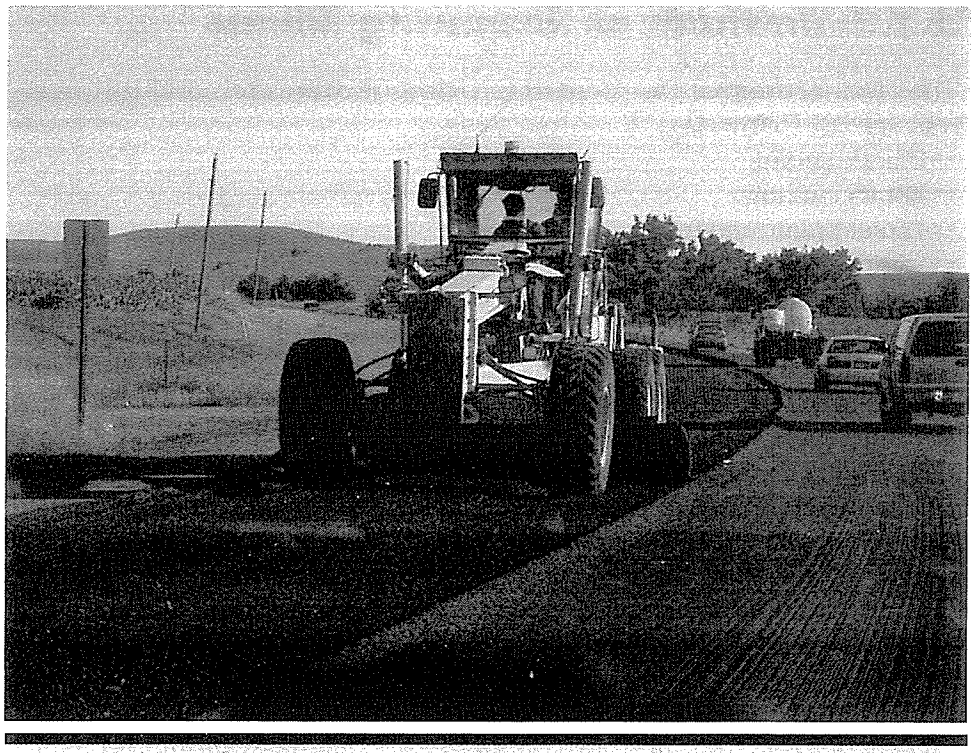


Figure 15.2 Blade Mixing

forth across the roadway with the blade until the aggregate particles are coated and the mix has a uniform appearance.

During mixing, attention should be paid to the blade's position. The blade should be positioned so that a complete rolling action of the material is obtained as the windrow is manipulated. Also during mixing, care must be taken not to incorporate extra material into the windrow and, at the same time, not to lose any material out of the windrow. After mixing and aeration have been completed, the windrow is placed to one side of the area to be surfaced in preparation for spreading.

Rotary Mixing

A rotary-type mixer consists of a mobile mixing chamber mounted on a self-propelled machine (see Figure 15.3). The chamber is open at the bottom and has widths ranging from 6 to 10 feet (1.8 to 3.0 meters). Inside the chamber are one or more transverse rotating drums that contain attached carbide-cutting teeth. This type of equipment is particularly useful for mixed-in-place cold recycling when old hot or cold mix surfaces are to be incorporated into the mix. The blades or cutting teeth serve a twofold purpose—cutting the in-place material to a specified depth and mixing it with asphalt. As the machine moves forward, it strikes off the freshly mixed material to a predetermined level. The asphalt can be introduced in one of two ways. In some mixers, the asphalt comes through a spray bar that extends inside the mixing chamber; the amount sprayed is governed by